

MooreVIEW Opta Firmware & Operation

Firmware version: 2.3.81

Document generated: 2026-08-13

Operator and integrator reference — MQTT Parc ST runtime on Arduino Opta

Contents

- [Overview](#)
 - [Device modes \(v2.3.46+\)](#)
- [Hardware](#)
- [Firmware architecture](#)
- [Network](#)
 - [Ethernet](#)
 - [MQTT broker](#)
 - [PC hub](#)
- [Web pages and APIs on device](#)
- [MQTT command reference](#)
- [NV storage](#)
- [ST program lifecycle](#)
- [Force and I/O](#)
- [Time sync](#)
- [Version history \(2.3.30+ highlights\)](#)
- [PC integration checklist](#)
- [Troubleshooting](#)
- [RS-485 fieldbus \(Modbus master / Pentair\)](#)

MooreVIEW Opta — Firmware & Operation Reference

Sketch: MooreviewOptaMqttSt

Current version: 2.3.81 (mv_version.h — sync from C:\data\MooreviewOptaMqttSt)

Protocol: MQTT Parc mooreview/v1/{deviceId}/... ; global P2P tags mooreview/v1/g/{siteKey4}/{tagName}

MooreVIEW PC integrates via driver type mqtt_parc . In-app help: **F1** → **MQTT Parc hub & Opta**.

Overview

MooreviewOptaMqttSt combines an on-device **ST bytecode runtime** with **MQTT Parc** telemetry and remote deploy. The PC hub ingests telemetry into data/parc.json ; MooreVIEW deploys programs with put_program , runs ST with runtime_start , and syncs forces and time over MQTT.

Device modes (v2.3.46+)

Mode	Opta	PC
Standalone (default)	Runs ST bytecode, full scan loop, NV auto-run	Enable Remote ST execution ; Download & Start deploys to Opta
Remote I/O	I/O scan only — reads inputs, writes outputs from PC via write_outputs ; rejects put_program	Remote ST execution OFF — PC scan engine runs ST; driver sends output values over MQTT

Set mode on /setup → **Device mode** , via MQTT set_device_mode , or save in NV config. Reboot after change.

Important: Parc deploy updates the ST program and NV storage only. It does **not** flash this sketch. After firmware changes in git, re-upload via Arduino IDE.

Hardware

Component	Details
Base	Arduino Opta — 8 digital inputs (I1–I8), 4 relays (R1–R4), raw analog I1_RAW–I8_RAW
Relay LEDs	On-board LEDs D0–D3 mirror relay states R1–R4
Expansion slot 1	Closest to base; up to 5 slots configured
AFX00005 (D1608E)	16 digital inputs + 8 relay outputs
AFX00007 (A0602)	8 analog channels + 4 PWM outputs
Identity	ATECC608B 9 bytes → FNV-1a 64 → deviceId = mv_ + 16 hex (legacy opta_{18hex} still accepted by PC registry)
Global site key	Admin-assigned uint16 (default 0x0001) for P2P globals; addr key = 4 lowercase hex digits
Network	Ethernet (primary); optional WiFi setup AP

Requires **24 V** on expansion modules for detection. Install **Arduino_Opta_Blueprint** for expansion scan.

Firmware architecture

Module	Role
MooreviewOptaMqttSt.ino	Main loop, HTTP routes, deferred NV auto-run
mv_mqtt.cpp	Parc MQTT client, command dispatch, telemetry
mv_st.cpp, mv_bc.cpp	ST bytecode VM (same AST as MooreVIEW PC)
mv_tags.cpp, mv_io.cpp, mv_expansions.cpp	Tag model, physical I/O, expansion tags
mv_program_store.cpp	QSPI NV program file, CRC, auto-run flag
mv_store.cpp, mv_setup_web.cpp	Device config KV (Ethernet, broker, expansions, global site key)

<code>mv_global_key.cpp</code>	Global site key → addr key + topic helper (mooreview/v1/g/{key}/{tag})
<code>mv_io_map.cpp</code>	/io-map web UI and JSON API
<code>mv_rtc.cpp</code>	Software wall clock + queued HAL RTC
<code>mv_identity.cpp</code>	ATECC608 → mv_{16hex} deviceId (FNV-1a 64)
<code>mv_http.cpp</code>	Native Ethernet HTTP server
<code>mv_ota.cpp</code> , <code>mv_version.cpp</code>	OTA upload, version reporting

Serial debug: USB **115200**. Milestone logs use `[MV*]` prefix (always on).

Network

Ethernet

DHCP or static IP configured on `/setup`. Default static example: `192.168.1.50`.

MQTT broker

- **Topic prefix:** `mooreview/v1`
- **Telemetry:** `mooreview/v1/{deviceId}/telemetry` (no ATECC serial in JSON)
- **Global tags (P2P):** `mooreview/v1/g/{siteKey4}/{tagName}` — site key from NV config (default `0001`); firmware v2.3.47+ pub/sub at telemetry rate (retained); PC hub mirrors into tag store
- **Commands:** `mooreview/v1/{deviceId}/cmd`
- **Responses:** `mooreview/v1/{deviceId}/cmd/response`
- **Config:** `mooreview/v1/{deviceId}/config` (pause telemetry during debug)

Broker IP/port saved in device NV on `/setup` (example: `192.168.1.233:1883`). Must match MooreVIEW **System setup** → **MQTT broker URL** (PC LAN IP, not `127.0.0.1` from the device). Reboot after broker change.

PC hub

Enable **MQTT Parc hub** and set broker URL in MooreVIEW System setup. Start Mosquitto on the PC LAN interface (`npm run mqtt:start`).

Web pages and APIs on device

Method	Path	Purpose
GET	/, /setup	Setup GUI — Ethernet, WiFi AP, MQTT broker, global site key , device mode , expansions, ST status
GET	/io-map	Live I/O map (base + expansions); Enable I/O update polls API
GET	/api/status	Full health JSON
GET	/api/status/lite	Lightweight status (setup panel refresh)
GET	/api/tags	Tag snapshot
GET	/api/io-map	I/O map JSON (logic vs effective when forced)
GET/PUT	/api/setup/config	Read/write persisted device config
POST	/api/setup/scan	Scan expansion modules
POST	/api/setup/reboot	Reboot device
PUT/DELETE	/api/program	Local AST deploy / clear (engineering)
GET/PUT	/api/program/autorun	NV auto-run flag
POST	/api/runtime/start , /api/runtime/stop	Local runtime control
GET	/api/ota	OTA status
POST	/api/firmware	Flash .bin (reboots)

Navigation bar links **Setup** ↔ **I/O Map** on both pages.

MQTT command reference

Publish JSON to `mooreview/v1/{deviceId}/cmd` :

```
{ "id": "<uuid>", "op": "<command>", "body": { ... } }
```

Command	Body (key fields)	Effect
<code>put_program</code>	<code>bc</code> , <code>tagIds</code> , <code>tags</code> , <code>protocolVersion</code> , <code>clientVersion</code> , optional <code>autoRunOnBoot</code>	Load bytecode, save to QSPI NV, register tags
<code>runtime_start</code>	<code>scanMs</code> (default 100)	Start ST scan loop
<code>runtime_stop</code>	—	Stop scan loop
<code>runtime_status</code>	—	Return running state, <code>programOk</code>
<code>set_force</code>	<code>tagId</code> , <code>forceInput</code> , <code>forceOutput</code> , <code>forceValue</code>	Apply PLC-style force
<code>clear_force</code>	<code>tagId</code>	Release force on tag
<code>sync_time</code>	<code>unixUtc</code> , <code>tzOffsetMin</code>	Set software wall clock (not during deploy)
<code>set_autorun</code>	<code>enabled</code> or <code>autoRunOnBoot</code>	Persist auto-run on power-up (KV)
<code>clear_program</code>	—	Clear program, stop runtime
<code>get_program</code>	—	NV status: <code>programFromNv</code> , <code>programNvCrc</code> , <code>autoRunOnBoot</code>
<code>scan_expansions</code>	—	Rescan modules, refresh expansion tags
<code>write_outputs</code>	<code>outputs</code> : { "R1": true, ... }	Set output tag values (Remote I/O mode; PC scan writes)
<code>set_device_mode</code>	<code>mode</code> : <code>standalone</code> <code>remote_io</code>	Persist device mode in NV (reboot recommended)

`put_program` blocks other commands until complete. In **Remote I/O** mode, `put_program`, `clear_program`, and `set_autorun` are rejected. Minimum firmware for Parc deploy: **v2.3.18+**. Recommended: **v2.3.41+** (reliable MQTT cmd subscribe).

NV storage

Store	Location	Contents
Device config KV	<code>/kv/mv_setup</code>	Ethernet, WiFi AP, expansion slot types, MQTT broker IP/port , device mode , globalSiteKey

Auto-run flag KV	/kv/mv_autorun	autoRunOnBoot (0/1)
Program file	QSPI FAT /fs/mv_program.bin	Header (name, CRC, auto-run) + bytecode

Program NV uses Modbus CRC16 over bytecode (matches PC `bcDeployCrc`). On boot, firmware loads NV program before MQTT; if `autoRunOnBoot` is set, ST starts ~500 ms after MQTT connects.

ST program lifecycle

```

Power-up → load NV program (if present) → Ethernet + MQTT connect
        → [autoRunOnBoot?] defer runtime_start ~500 ms
PC Connect → runtime_status + sync_time
PC Download & Start → put_program (skip if NV CRC matches) →
runtime_start
Deploy → save bytecode to QSPI (async write phases)
Stop → runtime_stop (program remains in NV)
clear_program / DELETE /api/program → clear NV program

```

MooreVIEW **System setup** → **Auto-run ST on Opta after power-up** sends `autoRunOnBoot: true` with deploy. PC boot auto-start waits ~8 s before deploy when Remote is on.

Force and I/O

- **Mux model:** `logicValue` = hardware read; effective tag getters return forced value when `forceInput` or `forceOutput` is set.
- **PC Tags** → **Force** sends `set_force` / `clear_force` over MQTT when Remote execution is on.
- **Relay outputs:** R1–R4 drive relays and mirror to LED_D0–LED_D3.
- **I/O map:** `/io-map` shows forced points highlighted; enable polling with **Enable I/O update**.
- **Expansions:** Tags auto-created after scan (e.g. expansion DI/DO/AI naming from slot index).

Time sync

- PC sends `sync_time` on **Connect** and **daily** for linked Opta drivers.
- Firmware sets an immediate **software wall clock** for timestamps and queues HAL RTC when idle (HAL write can block ~15 s — not done during deploy).
- `put_program` does **not** sync time.
- Before sync, serial logs use uptime; after sync, `YYYY-MM-DD HH:MM:SS` . Setup **Clock** row shows `rtcTime` .

Version history (2.3.30+ highlights)

Version	Notes
2.3.81	EZ Meter fieldbus (<code>MV_EZMETER</code>), MCSA M7/monitor (<code>mv_mcsa_*</code> , <code>/mcsa</code>), CT cal + edge AI updates
2.3.48	Phase 1 device id <code>mv_{16hex}</code> (FNV-1a 64 of ATECC 9 bytes); PC accepts legacy <code>opta_*</code>
2.3.47	Global site key in NV + <code>/setup</code> ; ATECC serial redacted from MQTT telemetry; <code>mv_global_key</code> topic helpers
2.3.46	Device mode: Standalone vs Remote I/O; <code>write_outputs</code> , <code>set_device_mode</code> ; <code>/setup mode</code> selector
2.3.45	Prior release
2.3.41+	MQTT cmd subscribe reliability; fix telemetry-OK-but-cmd-timeout
2.3.30– 2.3.40	NV program store, skip-deploy CRC, I/O map web UI, expansion telemetry
2.3.24+	Native HTTP, RTC software clock, broker NV on <code>/setup</code>
2.3.18+	Minimum for MQTT <code>put_program</code> deploy from MooreVIEW
2.3.11+	Fix 2.3.8 stack overflow in MQTT cmd handler
2.3.8	Do not use — ~20 KB stack buffer broke all MQTT

Always verify with GET `/api/status` → `firmwareVersion` .

PC integration checklist

- ☐ Flash **MooreviewOptaMqttSt v2.3.81+** via Arduino IDE
- ☐ Set MQTT broker on Opta `/setup` (LAN IP, port 1883) → Save → Reboot
- ☐ Set **Global site key** on Opta `/setup` and MooreVIEW System setup (`mqttParc.globalSiteKey` , default `1`)
- ☐ Choose **Device mode** on `/setup` : Standalone (ST on Opta) or Remote I/O (PC runs ST)
- ☐ Start Mosquitto on PC LAN (`npm run mqtt:start`)
- ☐ **System setup**: enable MQTT Parc hub, broker URL; **Remote ST execution** ON for Standalone, OFF for Remote I/O
- ☐ Add `mqtt_parc` driver (template or bulk-add from Parc registry); `deviceId` matches firmware
- ☐ **Sync tags from device** on driver card; **Scan expansions** if modules fitted
- ☐ Optional: **Auto-run ST on Opta after power-up** in System setup
- ☐ **Program** → **Remote** on → **Download & Start**
- ☐ Optional: assign **position ID** for plant location; use **Replace hardware** for change-outs
- ☐ Commissioning: **Tags** → **Force** (syncs to Opta); **Live I/O** → **Enable I/O update**
- ☐ Operations: **Report** → **MongoDB logs** for system log and hardware history

Troubleshooting

Symptom	Likely cause	Action
No telemetry	Broker/firewall/Mosquitto	Match PC + Opta broker IP; check LAN routing
Telemetry OK, cmd timeout	Old firmware or subscribe fail	Reflash v2.3.41+ ; Serial: MQTT subscribed cmd+config
Broker mismatch	<code>127.0.0.1</code> on device	Set PC LAN IP on <code>/setup</code> and System setup
Deploy fails / old fw	Firmware < 2.3.18	Arduino IDE upload MooreviewOptaMqttSt
HTTP dead during deploy	Large put on old fw	Use MQTT deploy; upgrade firmware
ST not running after link	Linked ≠ running	Download & Start , not Connect alone
Wrong deviceId	Manual id typo	Use Add from Parc registry or read <code>/setup</code>

NV program not auto-run	Flag off or no deploy	Deploy once; enable auto-run on PC and/or device
Clock shows uptime	No sync yet	Connect driver or wait for daily sync_time
Expansions missing	24 V / library	Seat modules in slot 1; install Opta Blueprint; Scan

Serial always-on: `[MV*]` boot, subscribe, cmd lines @ **115200**.

RS-485 fieldbus (Modbus master / Pentair)

Path	Role	Notes
PC <code>modbus_rtu</code>	PC is Modbus master	Polls Opta slave sketch (<code>opta_rtu_slave.json</code>) or other Modbus devices on USB/COM
PC <code>pentair_rs485</code>	PC Pentair master	UltraTemp/MasterTemp heat pump @ 9600 8N1; use <code>pollIntervalMs</code> (e.g. 300000)
PC <code>vgreen_epc</code>	PC Regal GEN3 master	SPECK BADU pump @ 19200; default <code>pollIntervalMs</code> 120000
Opta <code>mqtt_parc</code>	ST + I/O over MQTT	Primary integration — no RS-485 in default firmware
Opta <code>mv_fieldbus</code> + <code>MV_EZMETER</code>	Edge Modbus master → MQTT tags	<code>mv_fieldbus.cpp</code> / <code>mv_ezmeter.cpp</code> — <code>-DMV_FIELDBUS=1 -DMV_EZMETER=1</code> ; DDS_* tags; not simultaneous with Modbus slave on same UART
Opta True Echo sample	Radar bring-up	../arduino-opta-true-echo/ standalone sketch; PC template <code>apg_true_echo_rtu</code>

Single transceiver rule: one RS-485 port cannot be Modbus master and slave at the same time. Split buses (different COM ports / adapters) or use Opta edge polling + MQTT.

Wiring (recommended hybrid):

- Opta RS-485 A/B → Pentair heat pump comm terminals (9600 8N1, addr `0x70`).
- PC USB-RS485 → SPECK BADU pump (19200 8N1, slave 21).
- Opta Ethernet → MQTT Parc hub on PC (ST, remote I/O).

Activation: Hardware wizard → Modbus RTU group → *Pentair UltraTemp* template; set COM, `deviceAddr` 112 (`0x70`), `pollIntervalMs` 300000. Enable driver; tags `HP_*` update on

poll interval while global `scanMs` stays fast.

- Flash/build: [README.md](#)
- PC MQTT Parc: `docs/MQTT_PARC.md` , `docs/BASELINE_TEST.md`
- ST programs: `st/opta-mqtt/README.md`